

PHYS 580 Homework 7 (Chap.10)

All questions are worth 10 points, irrespective of complexity or length. Please submit all your solutions on Brightspace in PDF format, together with the source code (if any). Make sure to

- a) discuss the physics and your results, as appropriate and/or directed,
- b) include graphical output to illustrate the results,
- c) include the source codes of your programs (at least the critical parts thereof),
- d) describe briefly what your program does, and how it does it (algorithm), and
- e) state the nature of the numerical approximation you used and demonstrate how you know that the approximation (with the parameters you used) is adequate for the problem you used it for.

Note the errata pages for the textbook: <http://www.physics.purdue.edu/~hisao/book/www/errata.pdf>
which are explained further on <http://www.physics.purdue.edu/~hisao/book/www/errata.html>

1. Problem 10.5 (p.322)

Use the shooting method to study how the wave function for a particle-in-a-box depends on the magnitude of the potential outside the box V_0 . Examine the variation of ψ beyond the walls of the box and show that it decays exponentially with distance in this region. Study the decay length as a function of V_0 and compare the results for different energy levels. As the energy level approaches V_0 the decay length should become larger.

2. Problem 10.12 (p.333)

Employ the variational-Monte Carlo method to calculate the ground-state energy and wavefunction of the anharmonic oscillator whose potential is given by $V(x) = x^4$.

3. Write a matching method program to study the coupling between two one-dimensional quantum mechanical systems in neighboring square wells that are separated by a small, square barrier (cf. Figs. 10.11 and 10.12 of the textbook). In particular, observe how identical *unperturbed* states in each well get mixed due to being coupled through the finite barrier. Demonstrate numerically, for at least two different examples (such as the two ground states and then two excited states), that the initially equal energy levels split up. Namely, the parity even mixture moves down in energy, while the parity odd one moves up. This phenomenon is discussed in Chapter 10 of the book (p.318-320).